

Retrieval of *in-situ* Lake Bottom Sediment Cores from Schirmacher Oasis: A prelude to palaeoclimatic studies using lacustrine sedimentation in Antarctic environment

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ABSTRACT

Climatic changes have a significant effect on the glacial and periglacial regions adjacent to the large inland ice masses of Antarctica. A perspective on climatic variability can be obtained by the study of sedimentation within the lacustrine systems.

An attempt was made during the expedition to retrieve long, *in-situ* lake bottom sediment cores using an indigenously designed and fabricated sediment coring system developed by the authors at Antarctica Division, GSI, considering the field constraints and extreme antarctic climate.

INTRODUCTION

Schirmacher Oasis is situated in the peripheral region of central Dronning Maud Land, East Antarctica between 70°44'33" S – 70°46'30" S latitudes and 11°22'04" E – 11°54'00" E longitudes. The low lying hills of the oasis are bounded in south by the polar ice cap, ice-shelf lies to the north and entire area is interspersed with more than a hundred fresh water lakes (Fig. 1). The size of these lakes varies from a few hectares to a few km² and the maximum depth of water in them ranges from a few meters to about 150m. Depending upon their disposition, lakes in Schirmacher Oasis have been classified as proglacial, landlocked or inland and epishelf lakes (Ravindra et al., 2002). Majority of these lakes occupy ice-scoured basins and troughs (Fig.2) vacated by retreating glaciers, though some lakes follow well defined lineaments and show elements of structural control (Beg et al., 2006).

Modern glacial environments hold an essential key to our knowledge of present processes of erosion, transport and deposition, particularly within

glacial environments. Lacustrine system of Schirmacher Oasis holds a rich repository of information within the sedimentary layers at the lake bottom. The uninterrupted nature of sedimentation in lacustrine environments can be used to reconstruct the past climate spanning historical to geological time scales based on their sedimentological characteristics – including provenance, transport mechanisms, interacting tectonics, if any, drainage and biological content. High resolution palaeoclimatic data can be obtained from sediment cores using sedimentological, elemental geochemistry, diatoms, plants and animal microfossils, lithology and mineral magnetic properties.

PALEOCLIMATIC STUDIES: LAKE BOTTOM SEDIMENT CORES

In-situ lake bottom sediment cores are most significant tool in the palaeoclimatic studies. Indian endeavor of lake sediment coring was initiated during 16th Indian Antarctic expedition using Hydro-Bios gravity corer. Sediment samples including short cores of about 45 to 60 cm were collected from two lakes of Schirmacher Oasis. Sedimentological and mineralogical characteristics of the lacustrine sediments from Schirmacher were dealt by Sinha & Chatterjee (2000).

Palynological investigations of various polar surface deposits and lake sediment cores from Schirmacher Oasis have generated quite useful pollen data on the past polar climatic events and episodes. At present, understanding of many aspects remains poor. Three folds climatic oscillations (arid-warm, humid warm and more humid) are recorded from one meter long polar lake sediments from Zub and long lakes during late Holocene (8,000 years BP) as evidenced by the fluctuation of the values of long distance transported micro biota along with local moss spores and algal assemblages (Bera, 2004). Another 50 cm long sediment core from Zub Lake yielded thecamoebians (Mathur et al., 2006). Although more detailed study is required but it is possible to elucidate the past climatic oscillation in polar region through palynostratigraphy (Bera and Khandelwal, 2003; Bera, 2006) using well corroborated pollen diagram and pollen-spore deposition model. Limnology, Palynostratigraphy coupled with Geomorphology and Geochemical studies on deep lake sediment cores can provide better understanding of polar climate. Asthana et al.(2006) have initiated a multi-disciplinary and multi-institutional work utilizing lake bottom sediment cores to decipher sedimentation and palaeoclimatic history as revealed in polar lacustrine environments.

THE CORING DEVICE

Considering the working conditions and field constraints, an indigenous coring device (Fig.3) was designed and fabricated at Antarctica Division, GSI by the authors. It comprises sharp, stainless steel corer with a core catcher made up of bent acrylic sheet, 2m long acrylic sample tube(diameter - 50mm), outer stainless pipe that houses sample tube within it , perforated stainless pipes of 1 inch diameter, connecting sockets and a reducer fitted at the top of outer pipe. The components can be easily attached at the working site. Corer needs to be screwed at the bottom of outer stainless tube with acrylic sample tube placed inside (Fig. 3, B) and the reducer has to be screwed at the top end where 1 inch dia pipe can be fitted with the help of connecting sockets. The length of entire device can be increased depending upon the depth of the lake bottom by additional lengths of 1 inch dia pipes using sockets. The top end can be hammered to facilitate deeper penetration (Fig. 3, D) of the device within the lake sediment. Thus, the in-situ sediment column gets collected in the acrylic sample tube which can be retrieved after pulling up the coring device (Fig.3, E).

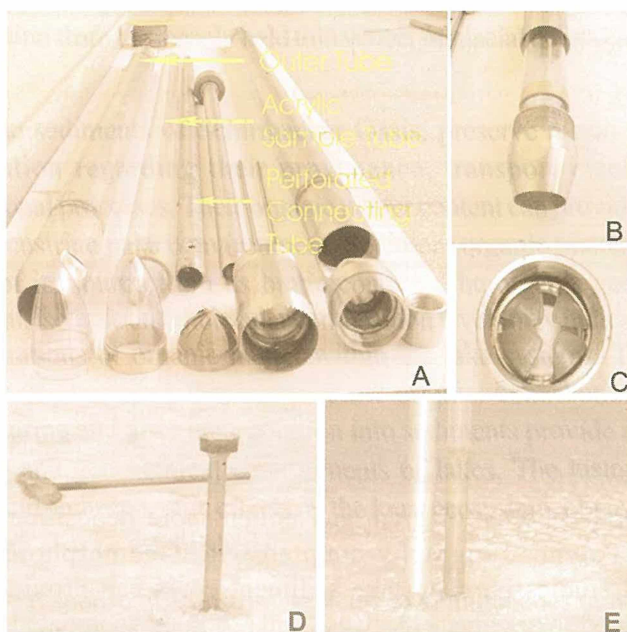


Fig. 3: A-Variety components of the Sediment coring device; B-The corer, Acrylic sample tube and outer tube. The corer along with core catcher and the sample tube can be screwed in the outer tube at the bottom portion; C- The acrylic core catcher housed within the corer; D-Coring device lowered below the frozen top lake surface for core retrieval from lake bottom; E-The outer tube and sample tube containing core sediments.

CORING AND RETRIEVAL OF LAKE BOTTOM SEDIMENT CORES

An attempt for retrieval of sediment core from Priyadarshini Lake was made on 25th May 2004. Suitable site was selected and a hole was drilled through the frozen lake with the help of a hand ice auger and crow bar. The thickness of the ice at the site of coring was 1.25m and the depth of the lake bottom from the surface was about 6m. Sediment column could not be obtained from this location as the corer had struck bouldery surface below and the tapering coring device was damaged due to striking against the rock surface. At another site on 27th May, 2004, the point near the junction of the reducer joining the corer and the connecting pipe broke while lowering the stainless steel corer. However, the corer was saved since it was attached to a wire line. No success could be achieved in subsequent attempts of coring in following months using the spare reducer after making some modifications in the core catcher. The temperature varied between -18°C and -22°C with wind velocity at its minimum. Because of the low temperature, the water in the hole would freeze within a few minutes of lowering the coring assembly making it necessary to use additional force for striking the corer for retrieving the sediment core adding to the risk of the corer snapping at the junction of the reducer and connecting pipe.

On October 16, 2004, during coring operations at Long Lake, the second reducer also broke from the same point i.e. the point near the junction of the reducer joining the stainless steel corer and the connecting stainless steel pipe. Lake sediment coring was resumed on 1st November 2004, using the welded reducer at the center in the SW part of Long Lake (Lat. S 70° 45. 381' & Long. E 11° 39. 585'). The thickness of ice at this site was about 1.65m and depth from the surface was around 5m. A sediment column of 90 cm was obtained from this point comprising predominantly dark coloured silty clay / sand / algal mats. The core sample was preserved in the freezer in an upright position for biological, sedimentological and mineralogical studies.

On 4th November, 2004 a few centimeters away from the above location, another hole was drilled (Lat. S 70° 45. 379' & Long. E 11° 39. 593') and 65 cm of sediment column was retrieved. Due care was taken to ensure that this sample is not exposed to light and the acrylic tube was removed from the corer in a dark room. The water content was siphoned out without disturbing the sediment column and wrapped the acrylic tube

with black polythene bags and secured with wide brown tape in the entire length of the acrylic tube after marking the top and bottom and kept for natural dehydration at room temperature. This sample will be used for dating by TL / OSL techniques.

Thus, two numbers of in-situ Lake Bottom Sediment cores were retrieved for detailed multidisciplinary palaeoclimatic studies.

THE CHARACTERISTICS AND SCOPE OF INVESTIGATIONS

The most characteristic feature of glacio-lacustrine environment is the deposition of debris supplied by ice-mass – i.e. settling of material from melting ice in the ablation zone. The retrieved core sediments are relatively fine grained because the water column generally remains stagnant or has a low flow velocity, so that even finest particles get settled. The factors responsible for the final depositional process in these lakes are settling out of suspension along with wind blown materials by surficial currents, rolling and traction populations through insignificant bottom currents and little contribution from the loosely held thin veneer of glacial debris on the adjoining slopes.

The sediments of Schirmacher Oasis, preserve a rich repository of information regarding their provenance, transport mechanisms and depositional processes. Their organic matter content can provide information about lacustrine palaeoenvironment. Sediment organic matter also yields a record of its source from its biotic content. The organic geochemistry at elemental, isotopic and molecular composition levels may indicate the sources and alterations of organic matter within the lake systems. The types and amounts of alterations that occur to the original contributions of organic matter during and after incorporation into sediments provide records of the physical and geochemical environments of lakes. The history of climate change and anthropogenic effects on the local ecosystem, obviously of recent times, can also be established.

The response of lacustrine systems to simultaneous climatic, tectonic and morphological forcing is poorly understood. Sedimentary records of lakes provide a complete and fairly high resolution palaeoclimatic data because climate is one of the key factors of lacustrine evolution. Smaller and geologically recent lacustrine systems like that of Schirmacher Oasis

originated after the Last Glacial Maximum ($18,000 \pm$ years) are very significant as they are very sensitive to even minor tectonic or climatic changes.

Sedimentological, mineralogical and geochemical data in conjunction with the biogenic contents from insitu lake bottom sediment cores can be effectively used in palaeoclimatic and palaeoenvironmental interpretation. As a result, limnological research in Antarctic environment has increased in the recent past with contributions from such lake rich regions as the Antarctic Peninsula, McMurdo Dry Valleys and East Antarctic lake-rich Oases: Vestfold Hills, Larsemann Hills, Bunger Hills, Rauer Islands and Windmill Islands.

Isotopic covariance trends obtained from the lake system may show changes rapidly from short-lived isotopically light periods (i.e. from seasonal to pluriannual) to longer steady-state periods of heavier isotope (i.e. from pluriannual to millennial). Thus, a comprehensive limnological research along with lacustrine sedimentation has great potential to establish reliable information on the palaeoclimatic history, their trend and possible climate change in future.

In a nut-shell, various elements of proposed palaeo-climatological investigations using these lake bottom sediment cores can be summarized below:

**Petrologic
Techniques**

Sedimentological parameters

**Chemical
Methods**

Stable Isotope studies- O, C, N & Trace Metals

Palaeontology

Study of Pollens, Diatoms, Micro flora & fauna

Geomorphology

Study of Landforms

CONCLUSION

In pursuance of the field season programme of GSI (2003 – 2004), lake sediment coring was taken up as part of winter programme during the 23rd Indian Antarctica Expedition. Despite initial problems related to extreme working conditions during the antarctic winter months, two in-situ lake bottom sediment cores of 90 cm and 65 cm lengths were retrieved from L – 27 (the so called, Long lake) for the purpose of a detailed multidisciplinary palaeoclimatic investigation. The indigenously designed sediment coring system was used in Antarctica for the retrieval of sediment core but further refinement in the materials used in fabrication and improvement in the coring device is possible for better results. A comprehensive multidisciplinary investigation including sedimentological, geochemical, biological, palynological, micropalaeontological and mineral magnetic susceptibility studies have been envisaged along with isotopic and thermo luminescence (TL) dating to elucidate various parameters.

ACKNOWLEDGEMENTS

The authors wish to express their gratitude to the Director General, GSI for giving them the opportunity to participate in the expedition. Maj. P. Anand Kumar, L.Nk. Ram Magan and fellow expedition members rendered crucial help in our attempts during the sediment coring. We also owe our sincere thanks to Shri Rasik Ravindra, the then Director, Antarctica Division, without whose moral support, guidance and encouragement, this task would not have been a success. NCAOR is thanked for providing necessary logistic facilities and over all support during the entire course of expedition.

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